



DCT430SD

USER'S MANUAL



CONDUCTIVITY METER WITH DATALOGGING SD CARD

Please read this manual carefully and thoroughly before using this product.

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INTRODUCTION

Thank you for purchasing General Tools & Instruments' DCT430SD Conductivity Meter with Datalogging SD Card. Please read this user's manual carefully and thoroughly before using the instrument.

The DCT430SD is a general-purpose handheld instrument suitable for measuring the conductivity of a solution or its total dissolved solids (TDS) or salt content. The meter also measures the solution's temperature.

The meter has four automatically switched full-scale ranges of conductivity measurement: 200 μS (microSiemens), 2 mS (milliSiemens), 20 mS and 200 mS. It also has four automatically switched ranges of TDS measurement: 200 ppm, 2,000 ppm, 20,000 ppm and 100,000 ppm. Salt content of up to 12% by weight also can be measured.

Among the applications suitable for the DCT430SD are water conditioning, wastewater monitoring, beverage production, aquaculture (fish farming), aquariums, pulp and paper processing, electroplating, and photography.

The DCT430SD has the performance and features needed to satisfy the most demanding aspects of the above applications. A front-panel pushbutton enables rapid switching among the three measurement modes. Measurement accuracies are $\pm 2\%$ of full-scale reading + 1 digit for conductivity and TDS; $\pm 0.8^\circ\text{C}/\pm 1.5^\circ\text{F}$ for temperature, and $\pm 0.5\%$ of the value for salt. The supplied carbon rod electrode conductivity/TDS/salt probe can be calibrated using standard solutions for any of three full-scale measurement ranges. All three standard solutions are available from General as optional accessories.

Because it is microprocessor-based, the DCT430SD can make full use of the portability, reliability and large storage capacities that SD memory cards offer. Measurements can be made automatically at any sampling rate between one second and nine hours. After time-stamping and storing the measurements on an SD card plugged into the instrument (a process called datalogging), the user can remove the card and plug it into a laptop or desktop computer either directly or via a USB card reader. The data logs are stored on the card as files with the .xls extension, which can be opened by Microsoft's Excel application.

The DCT430SD has a backlit 2-1/2 in. diagonal display and is powered by six "AA" Alkaline batteries or an optional 9-VDC AC adapter.

KEY FEATURES

- Measures any solution's conductivity, total dissolved solids (TDS) content, salt content, temperature
- Comes in hard carrying case with carbon rod conductivity/TDS/salt probe, 1.413 mS standard solution, 2 GB SD memory card
- Measures conductivity using four automatically switched full-scale ranges: 200 μ S (microSiemens), 2 mS (milliSiemens), 20 mS, 200 mS
- Measures TDS using four automatically switched full-scale ranges: 200 ppm, 2,000 ppm, 20,000 ppm, 100,000 ppm
- Measurement accuracy for conductivity, TDS: $\pm 2\%$ of full-scale reading + 1 digit
- Also measures solution's temperature and salt content of up to 12% by weight
- Temperature compensation: Automatic from 32° to 140°F (0° to 60°C), w/compensation factor manually variable from 0 to 5% per degree Celsius
- Displays maximum and minimum readings
- Performs real-time automatic datalogging at sampling time settable from 1 second to 9 hours
- Outputs Excel-compatible (.xls) data logs
- Accepts SD memory cards of up to 16 GB capacity
- Big (2.5 in. diagonal) front-panel green backlit LCD is easy to read
- Powered by six "AA" batteries or optional 9V AC/DC adapter

OPERATING INSTRUCTIONS

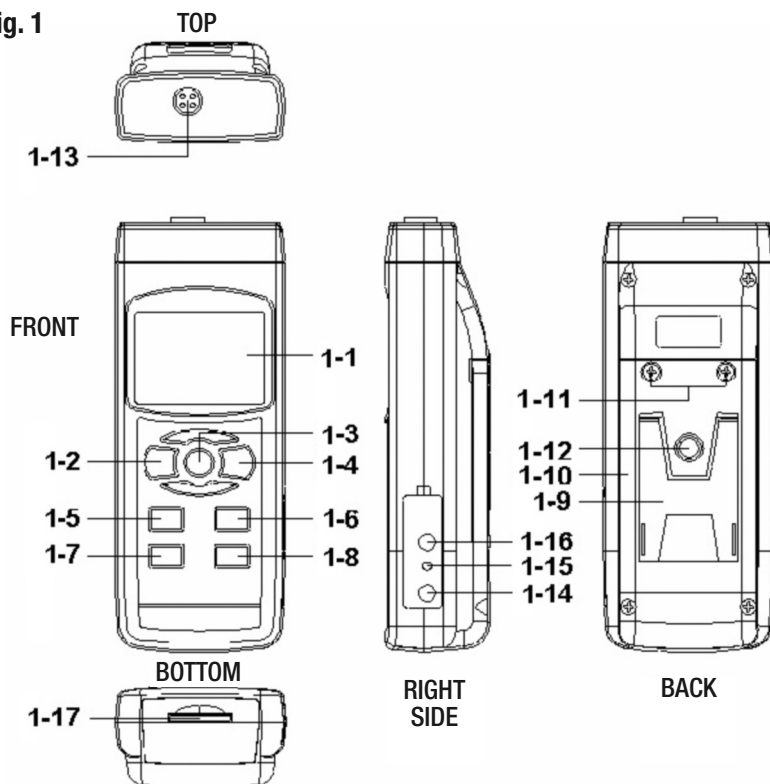
WHAT'S IN THE CASE

The DCT430SD comes fully assembled in a carrying case along with a conductivity/TDS/salt probe, a 2 GB SD memory card and this user's manual.

The only accessories for the meter available from General Tools & Instruments are standard conductivity solutions (1.413 mS, 12.88 mS, 80 μ S) needed to calibrate the probe and a 9VDC adapter for a 110V power supply. See the Optional Accessories section of this manual for more details.

Figure 1 shows all of the controls and indicators on the front, back, top, bottom and right side of the DCT430SD. Familiarize yourself with the positions and functions of these controls, indicators and connectors before moving on to the setup procedure.

Fig. 1



1. The DCT430SD's controls and indicators and other physical features

- | | |
|----------------------------|---|
| 1-1 Liquid-crystal display | 1-10 Battery compartment cover |
| 1-2 POWER button | 1-11 Battery compartment cover screws |
| 1-3 HOLD/ESC button | 1-12 Tripod attachment nut |
| 1-4 REC/ENTER button | 1-13 Conductivity/TDS/salt probe socket |
| 1-5 ▲ button | 1-14 Socket for 9VDC AC adapter |
| 1-6 ▼ button | 1-15 Reset button |
| 1-7 TIME button | 1-16 RS232 output jack |
| 1-8 LOGGER SET button | 1-17 SD card socket |
| 1-9 Kickstand | |

SETUP

1. Choose the power source. Before using the DCT430SD, be sure it is powered by fresh batteries or an optional 9VDC AC adapter plugged into the bottom jack on its right side (callout 1-14).

To remove the battery compartment cover, remove the two screws holding it in place (callout 1-11). Then install six “AA” batteries in the correct orientation, using the polarity marks on the inside of the compartment as a guide. Replace the cover by replacing the two screws.

2. Install an SD card. To prepare for setup, also install the supplied 2 GB SD memory card or another card with a capacity from 1 GB to 16 GB capacity in the socket on the bottom of the DCT430SD (callout 1-17). When installing the card, make sure its gold contacts are facing front and push the card into the socket until you hear a click. To remove the card, push it in until you hear a click and the card pops out.

Now power on the instrument by pressing the POWER button (callout 1-2) to generate a short beep. (To power off the DCT430SD, press the POWER button and hold it until the instrument responds with a long beep.)

When the meter powers on, a series of transient startup screens will briefly appear. Once the display has stabilized, perform the following eight setup steps in the order presented.

3. Format the SD card. The first step in the setup procedure determines whether or not to format the SD memory card plugged into the instrument. Press the LOGGER SET button (callout 1-8) and hold it for at least five seconds, until “Sd-F” appears in the lower half of the display. Quickly (within three seconds) press the REC/ENTER button.

(If you press buttons too slowly in setup mode, the screen will revert to the normal display. To return to the setup sequence, press and hold the LOGGER SET button again. To move ahead to the next field in the sequence or to the next parameter, press the HOLD/ESC button.)

Pressing the REC/ENTER button causes the word “no” or “yES” to appear over the term “Sd-F”. Press the ▼ button to make a selection. Choose “yES” whenever a new SD card is being used, or when a used card is being repurposed (from use with another SD card instrument or a camera, for example) and all data on it is to be erased. Choose “no” to preserve any data on a card previously used with this instrument.

If you choose “yES”, after you press the REC/ENTER button the instrument will prompt you to confirm that decision by displaying the term “Ent” below “yES” and sounding three beeps. To confirm that you want to begin the erasure/ formatting procedure, press the REC/ENTER. “Ent” will then flash several times and the instrument will sound another three beeps to confirm that the SD card has been erased and formatted.

4. Set the date and time. Once the SD card has been formatted (or not), the meter will automatically move to the next step in the startup sequence: setting the current date and time.

When the word “dAtE” appears in the center of the display, along with the flashing value “00.00.00” at the lower left, above “yy.mm.dd”, set the current year by pressing the ▼ or ▲ button repeatedly until the correct value appears above “yy”. Press the REC/ENTER button to store the setting. The next screen that appears will have the value above “mm” flashing. Use the ▼ or ▲ button to navigate to the current month and press the REC/ENTER button to store the setting. When the next screen flashes the value above “dd”, again use the ▼ or ▲ button to navigate to the current day and press the REC/ENTER button to store the setting.

Once you have set the date, the display will prompt you to set the hour, minute and second of the current time. Again use the ▼ or ▲ buttons to navigate to the correct values, and the REC/ENTER button to store the settings.

5. Set the datalogging sampling time. Once the date and time have been set and stored, the display will show a value below the letters “SP-t”. Press the ▼ or ▲ button to decrease or increase the flashing hour value until the desired number appears below “SP-t”. Press the REC/ENTER button to store that setting. The minutes value will then flash. Press the ▼ or ▲ button to decrease or increase the value and then press the REC/ENTER button to store that setting. Finally, press the ▼ or ▲ button to decrease or increase the flashing seconds value and then press the REC/ENTER button to store that setting. This completes the setting of the pressure meter’s datalogging sampling time.

6. Enable or disable auto power off. Once the sampling time has been set and stored, the display will show the word “yES” over the term “PoFF”. Press the ▼ button until the desired automatic power off management scheme (“yES” for enable; “no” for disable) is displayed. Press the REC/ENTER button to store the selection. If enabled, the power off function shuts off the DCT430SD after a period of inactivity of ten minutes.

7. Enable or disable the beeper. Once the auto power off function has been enabled or disabled, the display will show the word “yES” over the word “bEEP”. Press the ▼ button until the desired setting (“yES” or “no”) is displayed, and then press the REC/ENTER button to store the selection.

8. Choose a decimal point or comma to represent the decimal division between integers and fractions (for example, American-style 20.88 vs. European-style 20,88). Once the beeper has been enabled or disabled, the display will show the word “bASIC” above the phrase “dEC”. Press the ▼ button to make “bASIC” (American style) or “Euro” appear in the upper display, as desired. Press the REC/ENTER button to store the selection.

9. Select the temperature unit. Once the format of decimal point divisions has been set and stored, the lower portion of the display will show the term “t-CF”. Press either the ▼ or ▲ button to change the letter displayed at the right of the display to “C” (for Celsius units) or “F” (for Fahrenheit units), as desired. Then press the REC/ENTER button to store the selection.

10. Compensate the conductivity/TDS/salt probe for temperature. The default value is 2% per degree C. The value is adjustable from 0 to 5% per degree C. Once the default temperature unit has been selected and stored, the lower portion of the display will show the term “PER C”. Press the ▼ or ▲ button repeatedly until the value above “PER C” matches the desired temperature compensation level. Press the REC/ENTER button to store the setting.

The display will then either return to the first of the eight steps in the setup sequence—choosing whether or not to format the SD memory card—or show the term “ESC”. If “ESC” appears, press either the HOLD/ESC button or the LOGGER SET button to exit the setup procedure and enter normal operating mode.

CALIBRATING THE CONDUCTIVITY/TDS/SALT PROBE

The following procedure is sufficient to calibrate the supplied conductivity/TDS/salt probe so it accurately measures all three parameters.

1. Obtain the correct standard conductivity solution(s) for the full-scale range(s) you expect to use. For example, if you expect to measure conductivity values using a full-scale range of 2 mS, use a 1.413 mS standard conductivity solution. For measurements within the 200 uS full-scale range, obtain and use an 80 uS standard solution.

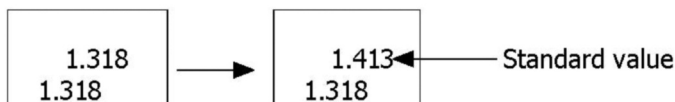
2. Plug the probe into its dedicated socket on the top of the meter (callout 1-15 of Fig. 1). Then power on the meter and push the ▲ button (which has the word “MODE” stenciled above it) until the display shows the term “Cd”.

3. To perform the calibration, immerse and shake the head of the probe in the standard solution. The display will show its conductivity in units of uS or mS

4. Once a value is displayed, use two fingers to press the REC/ENTER and HOLD/ESC buttons simultaneously.

Cd CAL

5. When this screen appears, release both fingers. Then press the REC/ENTER button. The display will then switch into a two-level (upper and lower) mode that looks like this:



6. Press the ▼ or ▲ button to raise or lower the measured value until it matches the standard conductivity value.

7. Complete the calibration procedure by pressing the REC/ENTER button.

If you expect to make conductivity readings using other full-scale ranges, you should repeat this calibration procedure using other standard conductivity solutions of appropriate strength. If you do, General advises performing the above procedure (for a 2-ms full-scale range) first.

NORMAL OPERATION

1. To prepare to make measurements, plug the connector at one end of the supplied conductivity probe into the socket on the top of the meter. Then power on the meter on by pressing the POWER button.

The DCT430SD2 has three operating modes. It can measure three properties of a solution: its conductivity, its total dissolved solids (TDS) content, and its salt content (by weight).

2. To select the parameter to be measured, press the ▲ button until the desired mode appears (as the term Cd, tdS or SALT).

Following are the default states for the DCT430SD:

- Conductivity measurement in units of microSiemens (uS) or milliSiemens (mS)
- Temperature measurement in °C
- Temperature compensation of 2% per °C
- Autoranging on and auto power off
- Sampling time of 2 seconds

3. To measure the conductivity of a solution, press the ▲ button until the term “Cd” appears. Hold the conductivity probe and immerse its head in the solution to be measured. Shake the probe to help any air bubbles inside it escape through the head.

The display will simultaneously show the solution’s conductivity (in units of uS or mS) in the center and the solution’s temperature (in oC) at the lower left.

As mentioned earlier, in the DCT430SD autoranging is enabled by default.

4. To manually choose a full-scale measurement range, press the ▲ button until the desired value (200 uS, 2 mS, 20 mS or 200 mS). Note that pressing the ▲ button also can re-enable autoranging. To change the default temperature unit or temperature compensation, perform Step 9 or 10 of the setup procedure.

If the display does not show a zero value before the probe head is immersed in the solution, you can quickly correct that error by doing the following. First, use the ▲ button to manually choose a full-scale measurement mode of 200 uS. Then press the ▲ button again, but hold it for at least ten seconds, until the display reads zero.

5. To measure the TDS content of a solution, press the ▲ button until the term “tdS” appears. Hold the conductivity probe and immerse its head in the solution to be measured. Shake the probe to help any air bubbles inside it escape through the head. The display will simultaneously show the solution’s TDS content (in units of ppm) in the center and the solution’s temperature (in °C) at the lower left.

6. To measure the salt content of a solution, press the ▲ button until the term “SALT” is displayed. Hold the conductivity probe and immerse its head in the solution to be measured. Shake the probe to help any air bubbles inside it escape through the head. The display will simultaneously show the solution’s salt content (by % weight) in the center and the solution’s temperature (in °C) at the lower left.

Whenever the DCT430SD is in measurement mode, you can turn off the backlight (which is on by default) by briefly pressing (but not holding) the POWER/ESC button. To reactivate the backlight, briefly press the button again.

Whenever the instrument is in measurement mode, you also can check the current date and time by briefly pressing the TIME button. Doing so causes both values to appear briefly at the lower left of the display.

HOLDING AND STORING MEASUREMENTS

7. To hold a measured value of pressure, press the HOLD/ESC button during the measurement. Doing so will cause the word “HOLD” to appear at the top of the display. Pressing the HOLD/ESC button again releases the hold.

8. To record and recall readings, press the REC/ENTER button while making measurements. This will make the term “REC” appear at the top of the display. Pressing the REC/ENTER button again, briefly, will make the term “MAX” appear to the right of “REC” and switch the display to the maximum value stored in memory during the last recording session. Pressing the REC/ENTER button again, briefly, will make the term “MIN” appear to the right of “REC” and switch the display to the minimum value stored during the last session.

9. To exit recording mode, press and hold the REC/ENTER button for at least three seconds, until the term “REC” disappears from the top line of the display. The display will then revert to showing the current reading.

AUTOMATIC VS. MANUAL DATALOGGING

The DCT430SD can automatically log data at a user-selected sampling period from 1 second to 9 hours. To view the sampling time for which the instrument was set up, press the **LOGGER SET** button (which has the words “Sampling check” stenciled below it) once. To change the sampling time, perform Step 5 of the setup procedure.

- 10. To start automatic datalogging**, press the **REC/ENTER** button once. The top line of the display will then show the term “REC”. Pressing the **LOGGER SET** button at this point will make REC flash and add the flashing term “LOGGER” at the top right of the display. This indicates that the instrument is currently storing measured values and their time stamps in memory.
- 11. To pause automatic datalogging**, press the **LOGGER SET** button once; this action makes the flashing term **LOGGER** disappear from the top right of the display and changes the term “REC” from flashing to constant. Pressing the **LOGGER SET** button again resumes automatic datalogging.
- 12. To end automatic datalogging**, press the **REC/ENTER** button and hold it for at least two seconds. This action causes the “REC” message to disappear.
- 13. To log data manually**, set the sampling time to zero using Step 5 of the setup procedure. Then press the **REC/ENTER** button once. The display will show the term “REC” on the top line, a value in the middle, and below it the letter “P” on the left and a number from 1 to 99 on the same line to the right. The number indicates the position on the SD card that will be used to store manually logged data.

Now press the **LOGGER SET** button. This will cause the beeper to sound and the term “**LOGGER**” to briefly appear at the upper right of the display. As in automatic datalogging mode, in this mode the instrument is storing measurements and their time stamps on the SD card.

In manual datalogging mode, however, measurements are being stored continuously (with a sampling time of zero), and their locations on the card can be changed.

- 14. To change the storage location of manually logged data**, press the ▼ button once; this causes the “P” to disappear from the left side of the display and the value on its line to begin flashing. Once the flashing begins, you can use the ▼ and ▲ buttons to change the flashing value to any number between 1 and 99. Once you have chosen the storage location, press the **REC/ENTER** button to save the setting. This causes the value to stop flashing and the “P” to return.
- 15. To end manual datalogging**, press the **REC/ENTER** button and hold it for at least three seconds. This action causes the “REC” message to disappear from the top line of the display.

TRANSFERRING DATA FROM THE SD CARD TO A COMPUTER

After automatic or manual datalogging of measurements, remove the SD card from the meter and plug it into your computer either directly (if it has an SD card slot) or through an SD card reader.

Because the files containing time-stamped data logs have the file extension .xls, they open in Microsoft's Excel application. Figures 2, 3 and 4 show three kinds of Excel presentations: a data-only screen, a graphics-only screen, and a mixed data/graphics screen.

	A	B	C	D	E	F	G
1	Position	Date	Time	Ch1_Value	Ch1_Unit	Ch2_Value	Ch2_unit
2	1	2009/8/12	13:12:36	73.1	uS	24.7	Degree_C
3	2	2009/8/12	13:12:38	73.2	uS	24.7	Degree_C
4	3	2009/8/12	13:12:40	73.2	uS	24.7	Degree_C
5	4	2009/8/12	13:12:42	73.2	uS	24.7	Degree_C
6	5	2009/8/12	13:12:44	73.4	uS	24.7	Degree_C
7	6	2009/8/12	13:12:46	73.5	uS	24.7	Degree_C
8	7	2009/8/12	13:12:48	73.4	uS	24.7	Degree_C
9	8	2009/8/12	13:12:50	73.4	uS	24.7	Degree_C
10	9	2009/8/12	13:12:52	73.5	uS	24.7	Degree_C
11	10	2009/8/12	13:12:54	73.3	uS	24.7	Degree_C
12	11	2009/8/12	13:12:56	73.2	uS	24.7	Degree_C
13	12	2009/8/12	13:12:58	73.2	uS	24.7	Degree_C

Fig. 2. Typical Excel data-only screen

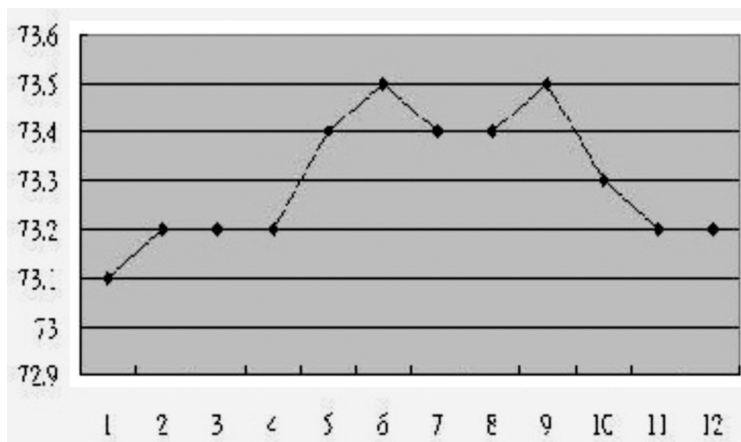


Fig. 3. Typical Excel graphics-only screen

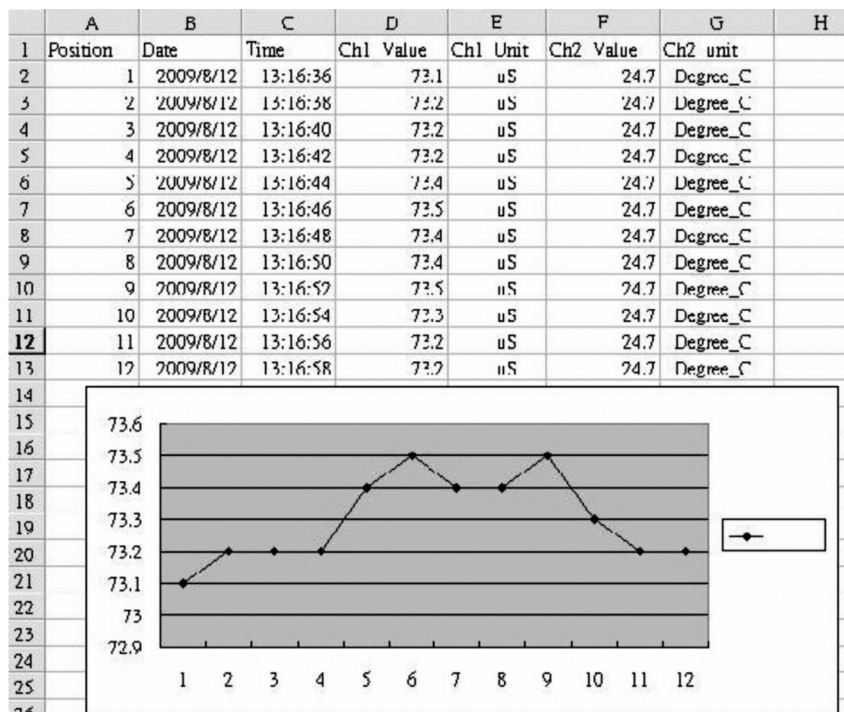


Fig. 4. Typical Excel mixed data/graphics screen

SPECIFICATIONS

Embedded microcontroller	Custom one-chip LSI device
Display type	Liquid-crystal with green backlight
Display size	2.05 x 1.5 in. (52 x 38 mm)
Probe type	Carbon rod electrode
Parameters (units) measured	Conductivity (uS, mS); total dissolved solids (ppm); salt content (% weight); temperature (°F, °C)
Measurement accuracy	± 2% of full-scale reading + 1 digit for conductivity, TDS; ± 0.8°C/± 1.5°F for temperature; ±0.5% of value for salt
Measurement resolution	Conductivity: 0.1 uS @ 200 uS full-scale (f.s.), 0.001 mS @ 2 mS f.s., 0.01 mS @ 20 mS f.s., 0.1mS @ 200 mS f.s.; TDS: 0.1 ppm @200 ppm f.s., 1 ppm @ 2,000 ppm f.s., 10 ppm @ 20,000 ppm f.s., 100 ppm @ 100,000 ppm f.s.; Temperature: 0.1°F/0.1°C; Salt: 0.01%

Temperature compensation	Automatic from 32° to 140°F (0° to 60°C); temperature compensation factor is manually variable from 0 to 5% per degree Celsius
Stored readings	Maximum, minimum
Datalogging sampling time	Settable between 1 second and 8 hours, 59 minutes, 59 seconds (in automatic datalogging mode)
SD card capacity	1 GB to 16 GB
Settable parameters	Date, time, auto power off, beep sound, sampling time, decimal point or comma decimal division, temperature unit, temperature compensation level
Operating temperature	32° to 122°F (0° to 50°C) for meter; 32° to 140°F (0° to 60°C) for probe
Operating relative humidity	0 to 85%
Power source	6 Alkaline “AA” batteries or optional 9-VDC AC adapter
Power consumption	14 mADC (normal operation, with backlight off and SD card not saving data); 37 mADC with backlight on and card saving data; 49 mADC with backlight on and card saving data
Dimensions of meter	6.97 x 2.68 x 1.77 in. (177 x 68 x 45 mm)
Weight of meter	1.08 lb. (489g)
Dimensions of probe	0.86 in. (diameter) x 4.72 in. (L) (22 mm x 120 mm)

MAINTENANCE & TROUBLESHOOTING TIPS

Keep the conductivity/TDS/salt probe clean and in the carrying case when not in use.

When the icon  appears in the left corner of the display, it's time to replace the six “AA” batteries that power the instrument (although measurements will remain valid for several hours after the low-battery indicator first appears). Replacing the batteries requires removing the two screws that hold the battery compartment cover in place, as explained earlier. After inserting fresh batteries in the correct orientation, tighten the screws to secure the cover.

If the meter “freezes” (like a computer) and buttons become unresponsive, try resetting the instrument by pushing the RESET button on its right side (callout 1-15 of Fig. 1) with the end of a paper clip.

Remove the batteries when storing the meter for an extended period of time.

Do not drop or disassemble the instrument.

OPTIONAL ACCESSORIES

- Standard conductivity solutions (1.413 mS, 12.88 mS, 80 uS)
- 9-VDC adapter for a 110VAC power supply

WARRANTY INFORMATION

General Tools & Instruments' (General's) DCT430SD Conductivity Meter with Datalogging SD Card is warranted to the original purchaser to be free from defects in material and workmanship. Subject to certain restrictions, General will repair or replace this instrument if, after examination, the company determines it to be defective in material or workmanship for a period of one year.

This warranty does not apply to damages that General determines to be from an attempted repair by non-authorized personnel or misuse, alterations, normal wear and tear, or accidental damage. The defective unit must be returned to General Tools & Instruments or to a General-authorized service center, freight prepaid and insured.

Acceptance of the exclusive repair and replacement remedies described herein is a condition of the contract for purchase of this product. In no event shall General be liable for any incidental, special, consequential or punitive damages, or for any cost, attorneys' fees, expenses, or losses alleged to be a consequence of any damage due to failure of, or defect in any product including, but not limited to, any claims for loss of profits.

RETURN FOR REPAIR POLICY

Every effort has been made to provide you with a reliable product of superior quality. However, in the event your instrument requires repair, please contact our Customer Service to obtain an RGA (Return Goods Authorization) number before forwarding the unit via prepaid freight to the attention of our Service Center at this address:

General Tools & Instruments
80 White Street
New York, NY 10013
212-431-6100

Remember to include a copy of your proof of purchase, your return address, and your phone number and/or e-mail address.



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